

Lotus Leaf Surface Sprint

Build a surface that sheds water and dirt the way a lotus leaf does.

Win condition: Best clean runoff and least residue after one redesign.

THE SCIENCE YOU ARE USING

Hydrophobic micro-texture. A lotus leaf is covered in tiny bumps coated in wax. Water cannot settle into the texture, so droplets bead up into near-spheres and roll off, carrying dirt with them. The effect comes from roughness plus a water-repelling coating, not from being smooth.

Roughness plus coating. You texture and coat safe surfaces, then race droplets down a standard ramp. The best surface sheds water fast and leaves the least residue. Copying the leaf on purpose is biomimicry in materials engineering.

HOW TO BUILD AND RUN IT

- 01 **Compare bare surfaces.** Drop water on smooth and rough surfaces. Watch which makes the droplet bead up.
- 02 **Texture and coat.** Add micro-texture and a water-repelling layer to your test surface.
- 03 **Add a dirt challenge.** Sprinkle pepper or cocoa as fake dirt. A good surface carries it off with the droplet.
- 04 **Race down the ramp.** Run droplets down the standard ramp. Time the runoff and check the residue left behind.
- 05 **Redesign for less residue.** Adjust texture or coating to shed cleaner, then re-run.

Safety: Work over a tray and wipe spills right away so the floor stays dry. Use only the washable dirt proxy (pepper or cocoa) and classroom-safe wax (wax paper, or a candle rubbed on cold and never lit). Brush off loose sandpaper grit. No goggles are needed for dropping water onto a tilted surface. Allergy: Cocoa allergy substitute: pepper or washable soil.

MATERIALS

Wax paper	shared
Cardstock	shared
Sandpaper strips	shared
Masking tape	shared
Pipettes	1 set per team (4; 3 mL graduated, reused)
Water	shared (tap)
Pepper or cocoa	residue (washable dirt proxy)
Candle or paraffin wax	shared (rubbed on cold to coat a surface)
Ramps	1 per team (4; on-hand rigid board at a fixed tilt)

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): _____

How we built our surface (texture + coating we used): _____

The ONE thing we changed for the redesign (texture, amount of wax, tilt, or material): _____

Runoff time: baseline _____ **after redesign** _____ **Residue left: baseline** _____ **after redesign**

DEFEND YOUR DESIGN

What evidence made you change your design? _____

If you ran it again, what is the ONE thing you would change? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Fast clean runoff	35
Least residue	25
Surface design explanation	20
Redesign gain	10
Craftsmanship	10
Total	100